

Review for Quiz 3 Math 2270

Ex What is a subspace of a vector space V .

Ex Let A be an $m \times n$ matrix.

Define $N(A)$ and show that it is a subspace of $\mathbb{R}^n$

Ex Find the null space of $\begin{bmatrix} 1 & 3 & -1 & 2 \\ 2 & 1 & 0 & 3 \\ 0 & 7 & -2 & 7 \end{bmatrix}$

Ex Show that $\left\langle \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \middle| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\rangle$
equals $\left\langle \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \middle| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\rangle$

Ex Show that the set of polynomials satisfying $\frac{d^3}{dx^3}(p(x)) = 0$ is a subspace of $\mathbb{R}[x]$

Ex 10 pg 128, 17 pg 129, 18 pg 129

Ex If $U_1 \subset V$ and $U_2 \subset V$ are subspaces, show that $U_1 \cap U_2 \subset V$ is a subspace

Ex Construct a matrix with nullspace $\langle (1, 1, 1) \rangle$ (resp. with nullspace $\langle (1, 0, 1), (1, 1, 0) \rangle$)

Ex Show that $N(A \cdot B) \supset N(B)$, give an example where $N(A \cdot B) \neq N(A)$ and where $N(A \cdot B) \neq N(B)$.